Race, explain and transfer

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Evaluate a model against measured performance and defend design decisions.

SUCCESS CRITERIA - I can...

1. Report measured distance, time and average speed.
2. Compare actual speed with a stated prediction.
3. Use evidence to explain three physical design choices.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 60-tooth driver turns a 480-tooth gear. Give ideal output/input torque and speed ratios and state the loss assumption.

Marzano check - circle: ① ② ③

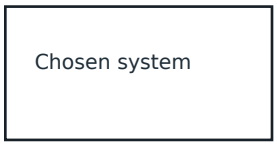
02 Worked example: explain the model

Reference: average speed = distance/time
relative deviation = |measured - predicted|/|predicted| × 100%. Prediction must be nonzero. Record real race conditions; this compares with a prediction, not symmetric percent difference.

Worked reasoning: A car covers 3 m in 2 s: average speed 1.5 m/s. A complete explanation also uses its circuit, gearing, friction and evidence from repeated runs; one race time does not isolate a cause.

Your turn: Draw an energy-flow diagram from battery to motion and other transfers; annotate it with one actual result, clearly labeled as measured.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw or annotate your final car. Label three choices involving wheels, gears, alignment, wiring or mass, then connect each to a physics idea.



Marzano check - circle: ① ② ③

04 Apply the physics

A supplied car covers 6.0 m in 8.0 s; its predicted speed was 0.90 m/s. Complete the table, find measured speed and percent deviation from prediction. Give a plausible explanation and identify a measurement that could test it.

Distance (m)	Time (s)	Measured v	Predicted v
___	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Use your actual race record to defend one successful design choice and identify one revision for a future build. Cite data or a specific observation for each.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A motor is 64% efficient and the following generator is 72% efficient. Find the combined useful fraction in sequence.
Label input, useful output and other energy transfers.

Draw and label all energy transfers across the boundary.

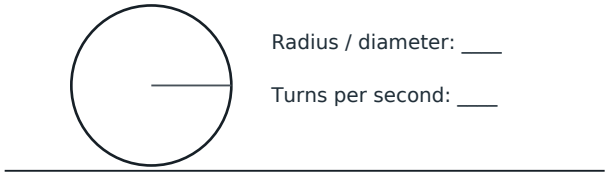
Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Transfer: a similar car uses twice the wheel diameter with the same loaded wheel rpm. Predict ideal speed change, then explain why testing is still necessary.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Supplied input is 6 V×1 A for 4 s. A 0.5 kg car reaches 2 m/s on a level track from rest. Find input energy and final translational kinetic energy. Explain why their ratio alone is not total drivetrain efficiency.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3